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Liaw

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(54) **PNEUMATIC HAMMER**

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B25D 17/04 (2006.01)

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CPC ***B25B 27/026*** (2013.01); ***B25B 27/0035*** (2013.01); *B25D 9/04* (2013.01); *B25D 17/043* (2013.01); *B25D 2250/121* (2013.01)

(58) **Field of Classification Search**

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See application file for complete search history.

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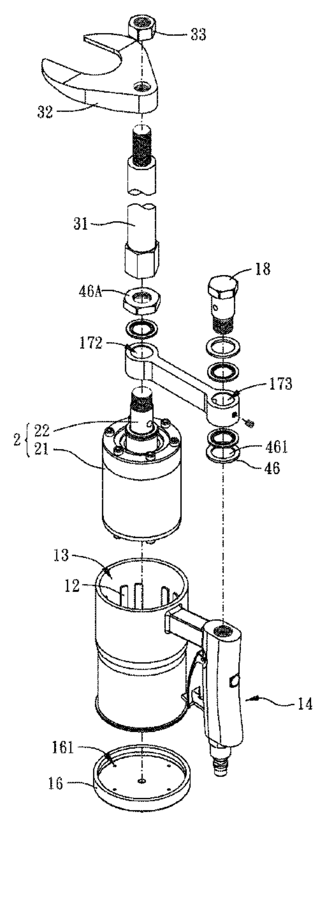
(57) **ABSTRACT**

A pneumatic hammer is provided, including: a handle assembly including a main body and a grip portion, the grip portion being connected to the main body and configured to be connected to a gas source, the main body including an interior space; a hammer assembly connected to and in communication with the grip portion, and including a hammer body and a shaft, the hammer body being movably connected to the shaft, and the hammer body being movably disposed in the interior space to hammer the shaft; and a working assembly including a rigid rod and a driving member, the rigid rod being connected to and movable with the shaft, the driving member being positioned to the rigid rod.

9 Claims, 6 Drawing Sheets



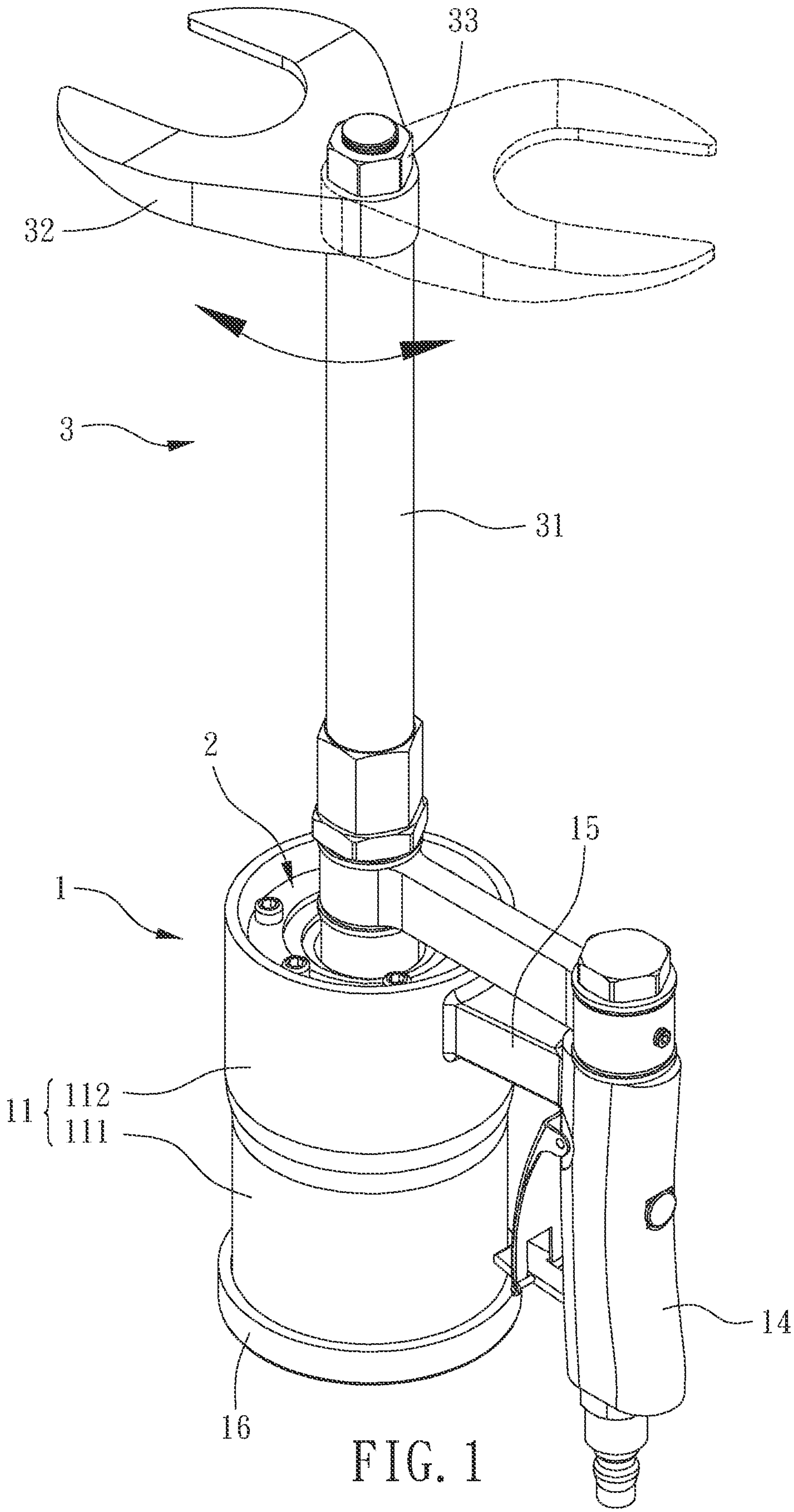


FIG. 1

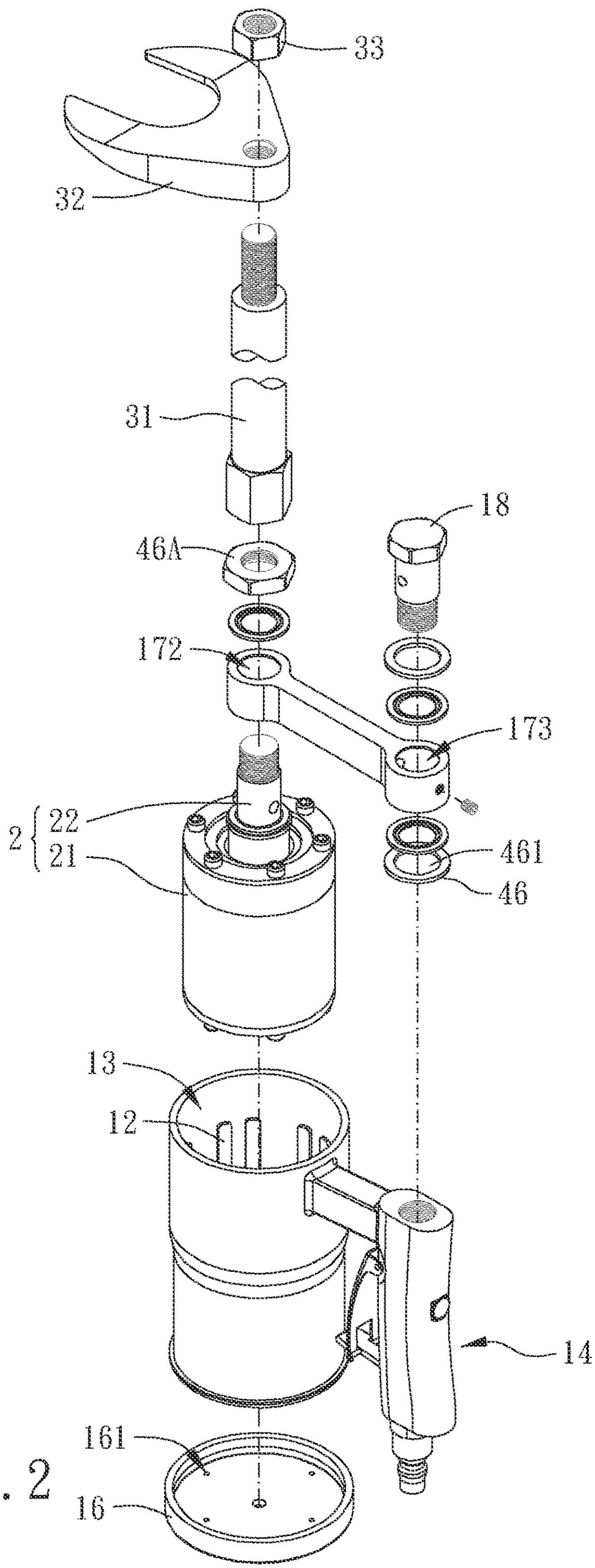


FIG. 2

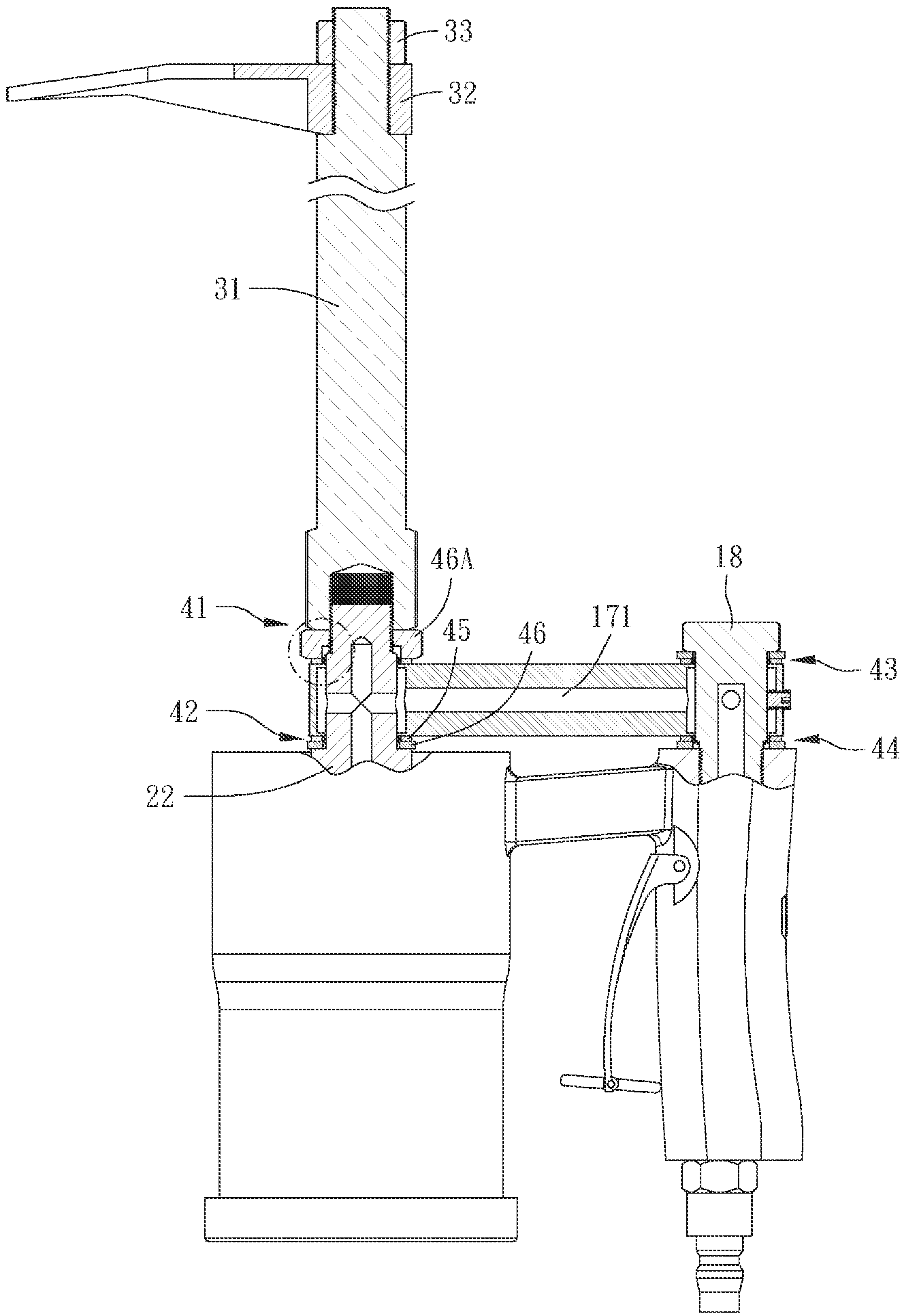


FIG. 3

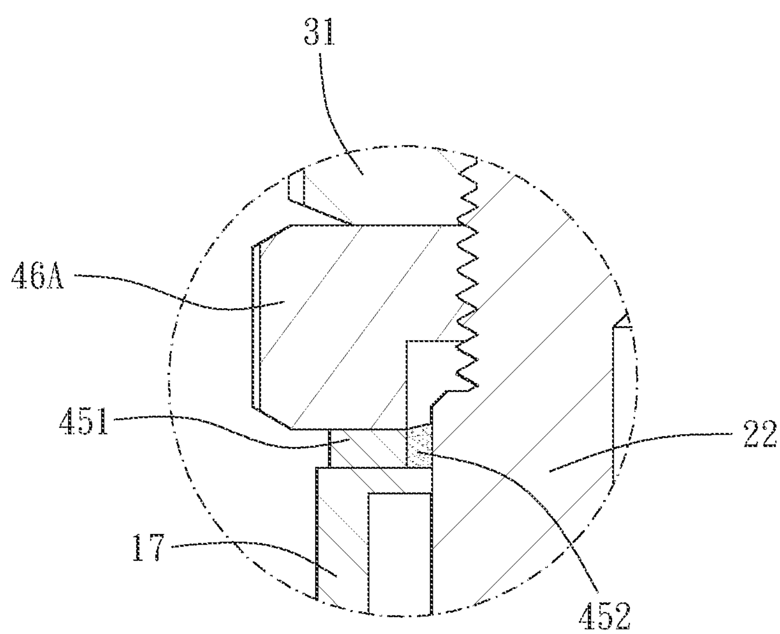


FIG. 4

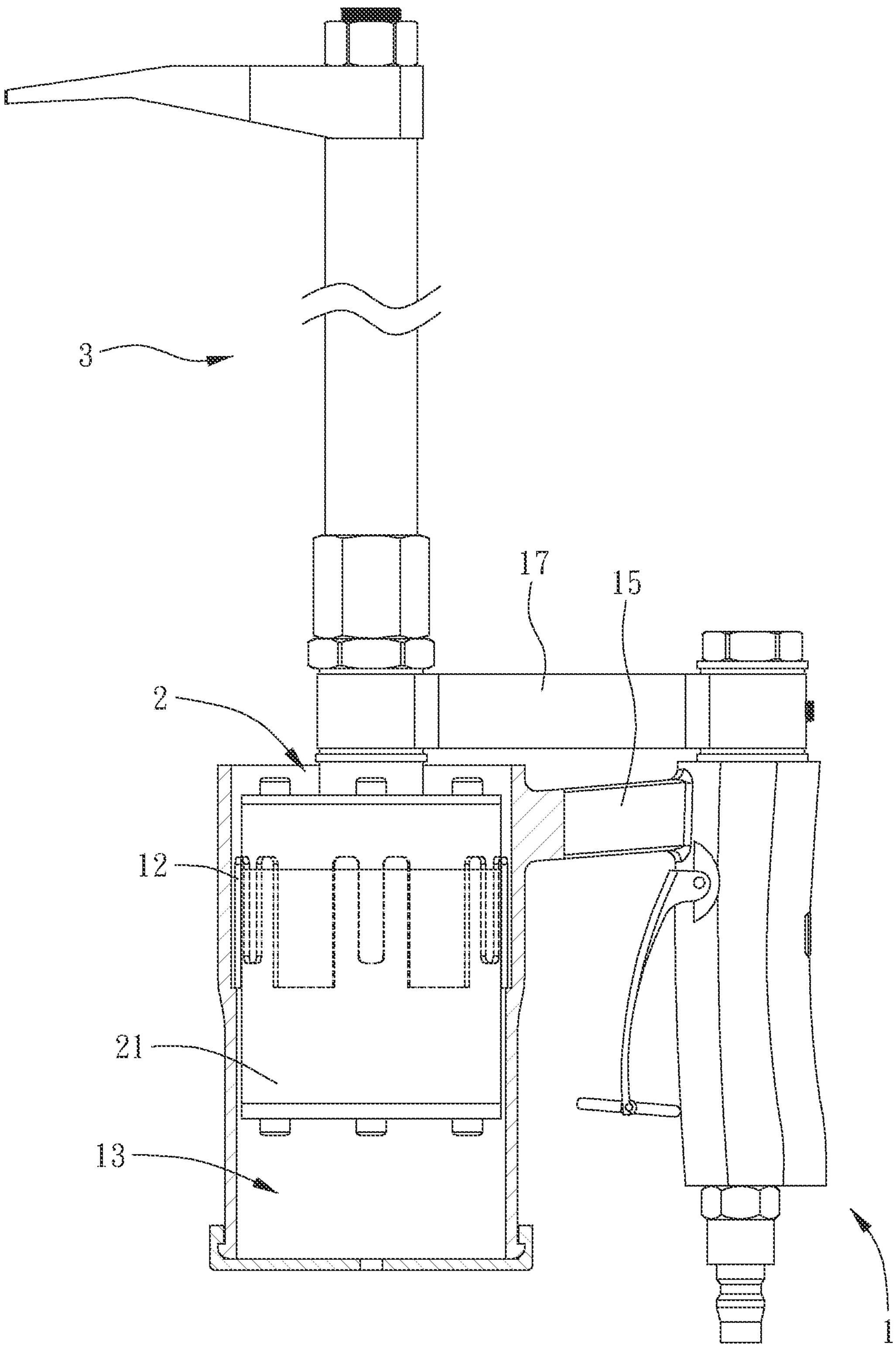


FIG. 5

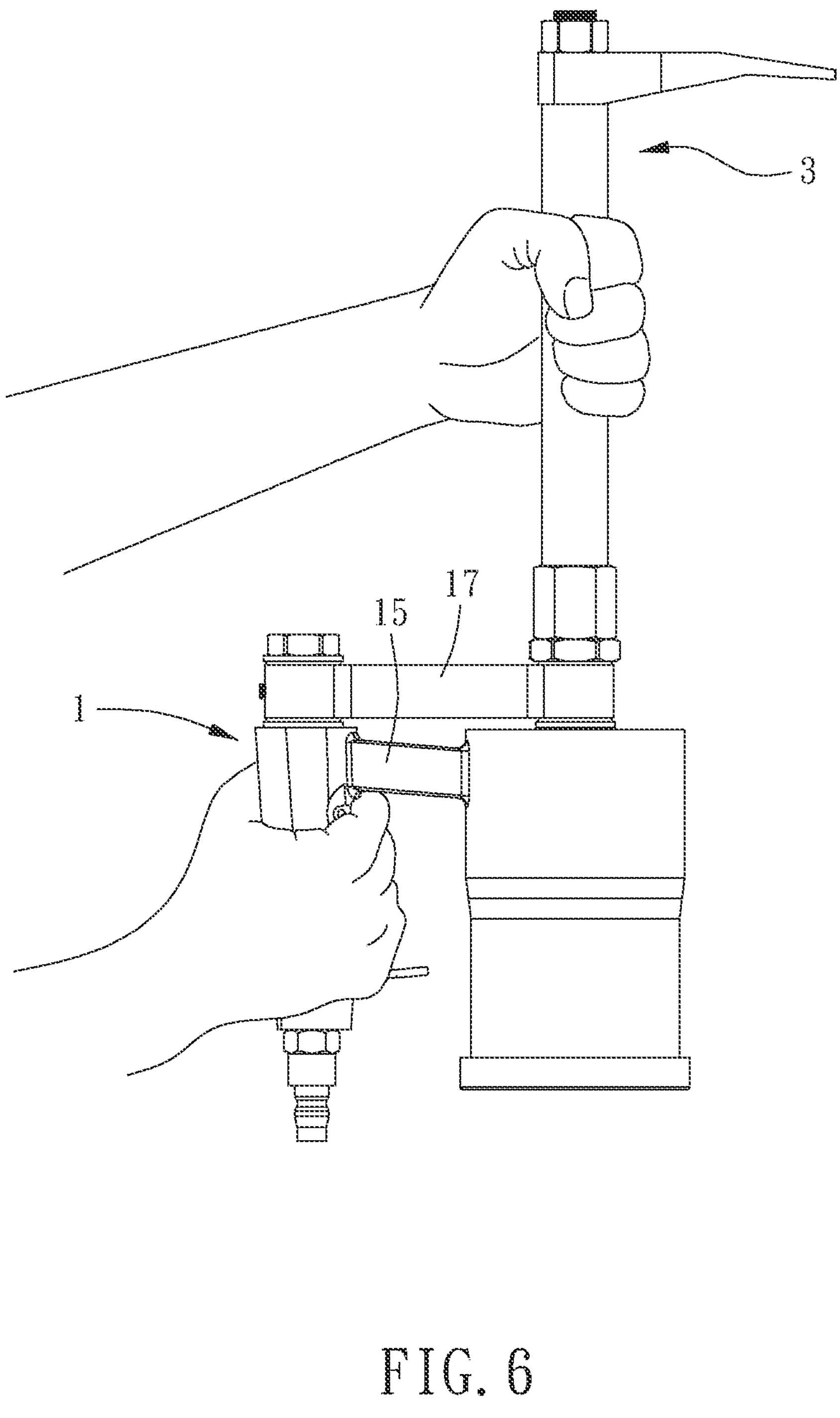


FIG. 6

PNEUMATIC HAMMER

BACKGROUND OF THE INVENTION

Field of the Invention

The present invention relates to a dismantling hand tool, particularly to a pneumatic hammer for dismantling.

Description of the Prior Art

Pneumatic hammer is a tool that uses compressed air as a power source. By driving the hammer to impact the shaft back and forth, the shaft transmits the impact force to the target to perform various operations such as pulling, hammering, digging or cutting. For use in maintenance of automobiles, a pneumatic hammer is often used to disassemble auto parts (such as fuel injectors) that are difficult to dismantle. This kind of tool is like one disclosed in TW 1595980 or U.S. Pat. No. 6,978,527.

However, the pneumatic hammer of TW 1595980 provides an insufficient impact force, is not ergonomic in operation, and needs to be assembled with various components, which is inconvenient and time-consuming to use. As to the pneumatic hammer of U.S. Pat. No. 6,978,527, there is a problem that the pneumatic hammer will move around uncontrollably due to the continuous action of the hammer unexpectedly disengaged from the fuel nozzle when the fuel nozzle is detached from the vehicle engine, and thus the pneumatic hammer can injure the user and cause damage to other parts of the vehicle.

The present invention is, therefore, arisen to obviate or at least mitigate the above-mentioned disadvantages.

SUMMARY OF THE INVENTION

The main object of the present invention is to provide a pneumatic hammer which can be stably and effectively gripped with high safety.

To achieve the above and other objects, a pneumatic hammer is provided, including: a handle assembly including a main body and a grip portion, the grip portion being connected to the main body and configured to be connected to a gas source, the main body including an interior space; a hammer assembly connected to and in communication with the grip portion, and including a hammer body and a shaft, the hammer body being movably connected to the shaft, and the hammer body being movably disposed in the interior space to hammer the shaft; and a working assembly including a rigid rod and a driving member, the rigid rod being connected to and movable with the shaft, the driving member being positioned to the rigid rod.

The present invention will become more obvious from the following description when taken in connection with the accompanying drawings, which show, for purpose of illustrations only, the preferred embodiment(s) in accordance with the present invention.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a stereogram of an exemplary embodiment of the present invention;

FIG. **2** is a breakdown drawing of FIG. **1**;

FIG. **3** is a side view with a partial cross-sectional portion of an exemplary embodiment of the present invention;

FIG. **4** is a partial enlargement of FIG. **3**;

FIG. **5** is a side view with another partial cross-sectional portion of an exemplary embodiment of the present invention;

FIG. **6** is a drawing showing operation of a pneumatic hammer of an exemplary embodiment of the present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

Please refer to FIGS. **1** to **6** for an exemplary embodiment of the present invention. A pneumatic hammer of the present invention includes a handle assembly **1**, a hammer assembly **2** and a working assembly **3**.

The handle assembly **1** includes a main body **11** and a grip portion **14**, the grip portion **14** is connected to the main body **11** and configured to be connected to a gas source, and the main body **11** includes an interior space **13**. The hammer assembly **2** is connected to and in communication with the grip portion **14**, the hammer assembly **2** includes a hammer body **21** and a shaft **22**, the hammer body **21** is movably connected to the shaft **22**, and the hammer body **21** is movably disposed in the interior space **13** to hammer the shaft **22** to move. The working assembly **3** includes a rigid rod **31** and a driving member **32**, the rigid rod **31** is connected to and movable with the shaft **22**, and the driving member **32** is positioned to the rigid rod **31**. The driving member **32** is horseshoe-shaped and can be hooked on objects, such as fuel injectors, to pull out the objects. The rigid rod **31** may be formed of one piece or multi-piece assembly.

Preferably, the rigid rod **31** is a solid rod so that it is sufficiently hard and strong to withstand the impact action force during operation.

Preferably, the rigid rod **31** has a length larger than a length of larger than the main body **11**, which is advantageous for using in narrow spaces.

In operation, the grip portion **14** can be held by a hand of the user, and the rigid rod **31** can be further held by the other hand of the user. In other words, the rigid rod **31** can be used as an auxiliary grip rod in addition to transmitting force. Additionally, the main body **11** can prevent the hammer body **21** from doing harm to people and objects due to unexpected impact of the hammer body **21**, which greatly improve safety.

Specifically, a plurality of ribs **12** project on an inner surface of the main body **11**, the hammer body **21** is slidably abutted against the plurality of ribs **12** so that the hammer body **21** and the inner surface of the main body **11** are gapped. As such, there is less contact area between the plurality of ribs **12** and the inner surface of the main body **11** so that it can effectively reduce contact friction resistance and ensure the smooth movement of the hammer body **21**.

The main body **11** is tubular, two ends of the main body **11** are in communication with outside the main body **11**, the main body **11** includes a first section **111** and a second section **112** connected to each other, the first section **111** and the second section **112** define the interior space **13**, and the second section **112** has an inner diametric dimension larger than an inner diametric dimension of the first section **111**. The plurality of ribs **12** project on the second section **112**, and the plurality of ribs **12** increase the structural strength of the second section **112**. When the hammer body **21** is moving in the interior space **13**, gas driven by the hammer body **21** enters into or exits from the interior space **13** with less drawing resistance, thus ensuring the stability of the hammer body **21** in movement.

Specifically, the second section **112** has an outer diametric dimension larger than an outer diametric dimension of larger than the first section **111**, and the first section **111** and the second section **112** form a stepped structure for helping to recognize the direction of assembly during the production assembly process. The first section **111** which has a smaller diametric dimension than that of the second section **112** facilitates gripping thereon and can be used as another auxiliary grip.

The handle assembly **1** further includes a linkage **15** and a lid **16**. The linkage **15** is laterally connected to and between the main body **11** and the grip portion **14** so that the main body **11** and the grip portion **14** are distanced for easy gripping. Preferably, the main body **11**, the grip portion **14** and the linkage **15** are integrally formed of one piece so that it has good structural strength. The lid **16** covers an end of the main body remote from the working assembly **3**, the lid **16** includes a plurality of through holes **161**, and the plurality of through holes **161** are in communication with the interior space **13**. The lid **16** can effectively prevent foreign matters (such as dust, fingers or the like) from entering the interior space **13**, and can protect personal safety. The lid **16** is preferably made of rubber, which is deformable according to the increase of gas pressure in the interior space **13**.

The handle assembly **1** further includes a rigid tube **17** and a connection rod **18**, the rigid tube **17** includes a channel **171**, a first connecting hole **172** and a second connecting hole **173**, the shaft **22** is disposed through the first connecting hole **172** and connected to the rigid rod **31**, the connection rod **18** is disposed through the second connecting hole **173** and connected to the grip portion **14**, and the shaft **22**, the channel **171**, the connection rod **18** and the grip portion **14** are in communication with one another for allowing the gas to flow in the hammer body **21**.

The pneumatic hammer further includes four seal assemblies for avoiding gas leakage, and each of the four seal assemblies includes a seal ring **45** and an annular member **46**. The seal ring **45** includes an outer ring body **451** and an inner ring body **452** which are connected to each other and concentrically arranged. The outer ring body **451** is made of rigid material, the inner ring body **452** is made of elastic material, and the inner ring body **452** is exposed within a central hole **461** of the annular member **46**. When the seal ring **45** and the annular member **46** are abutted against each other, a portion of the inner ring body **452** protruding beyond the central hole **461** is not pressed by the annular member **46** and can provide good sealing effect.

Specifically, the four seal assemblies include a first seal assembly **41**, a second seal assembly **42**, a third seal assembly **43** and a fourth seal assembly **44**. The first seal assembly **41** and the second seal assembly **42** are located at two sides of the first connecting hole **172**, the inner ring body **452** of the first seal assembly **41** and the inner ring body **452** of the second seal assembly **42** are disposed around the shaft **22**, the outer ring body **451** of the first seal assembly **41** is disposed and clamped between the rigid rod **31** and the rigid tube **17**, the outer ring body **451** of the second seal assembly **42** is disposed and clamped between the shaft **22** and the rigid tube **17**, the third seal assembly **43** and the fourth seal assembly **44** are located at two sides of the second connecting hole **173**, the inner ring body **452** of the third seal assembly **43** and the inner ring body **452** of the fourth seal assembly **44** are disposed around the connection rod **18**, the outer ring body **451** of the third seal assembly **43** is disposed and clamped between the connection rod **18** and the rigid tube **17**, and the outer ring body **451** of the fourth seal assembly **44** is disposed and clamped between the grip portion **14** and the rigid tube **17**.

Specifically, the seal ring **45** of the four seal assemblies have the same structure, the annular member **46** of the second seal assembly **42**, and the annular member **46** of the third seal assembly **43** and the annular member **46** of the fourth seal assembly **44** have the same structure. The thickness and rigidity of the annular member **46**A of the first seal assembly **41** are larger than the thickness and rigidity of the annular member **46** of the second seal assembly **42**, so that it can withstand greater impact force without deformation, to provide good structural integrity, airtight effect and service life.

Preferably, the shaft **22** is rotatably disposed through the first connecting hole **172**, and the rigid rod **31** is rotatable with the shaft **22**, for easy angular adjustment of the driving member **32**.

Preferably, the working assembly **3** further includes a nut **33**, the driving member **32** is detachably screwed to the rigid rod **31**, and the nut **33** is screwed to the rigid rod **31** and abutted against the driving member **32**. The nut **33** can ensure that the driving member **32** cannot rotate freely and detach from the rigid rod **31**.

Although particular embodiments of the invention have been described in detail for purposes of illustration, various modifications and enhancements may be made without departing from the spirit and scope of the invention. Accordingly, the invention is not to be limited except as by the appended claims.

What is claimed is:

1. A pneumatic hammer, comprising:

a handle assembly comprising a main body and a grip portion, the grip portion being connected to the main body and configured to be connected to a gas source, the main body comprising an interior space;

a hammer assembly connected to and in communication with the grip portion, and comprising a hammer body and a shaft, the hammer body being movably connected to the shaft, and the hammer body being movably disposed in the interior space to hammer the shaft; and

a working assembly comprising a rigid rod and a driving member, the rigid rod being connected to and movable with the shaft, the driving member being positioned to the rigid rod;

wherein the handle assembly further comprises a rigid tube and a connection rod, the rigid tube comprises a channel, a first connecting hole and a second connecting hole, the shaft is disposed through the first connecting hole and connected to the rigid rod, the connection rod is disposed through the second connecting hole and connected to the grip portion, and the shaft, the channel, the connection rod and the grip portion are in communication with one another.

2. The pneumatic hammer of claim **1**, wherein further including four seal assemblies, each of the four seal assemblies includes a seal ring and an annular member, the seal ring includes an outer ring body and an inner ring body which are connected to each other and concentrically arranged, the outer ring body is made of rigid material, the inner ring body is made of elastic material, the inner ring body is exposed within a central hole of the annular member; the four seal assemblies include a first seal assembly, a second seal assembly, a third seal assembly and a fourth seal assembly, the first seal assembly and the second seal assembly are located at two sides of the first connecting hole, the inner ring body of the first seal assembly and the inner ring body of the second seal assembly are disposed around the shaft, the outer ring body of the first seal assembly is disposed between the rigid rod and the rigid tube, the outer ring body of the second seal assembly is disposed between the shaft and the rigid tube, the third seal assembly and the fourth seal assembly are located at two sides of the second connecting hole, the inner ring body of the third seal assembly and the inner ring body of the fourth seal assembly are disposed around the connection rod, the outer ring body of the third seal assembly is disposed between the connection rod and the rigid tube, and the outer ring body of the fourth seal assembly is disposed between the grip portion and the rigid tube.

3. The pneumatic hammer of claim **2**, wherein the seal rings of the four seal assemblies have the same structure, the annular member of the second seal assembly, the annular member of the third seal assembly and the annular member of the fourth seal assembly have the same structure, and the thickness and rigidity of the annular member of the first seal assembly are larger than the thickness and rigidity of the annular member of the second seal assembly.

4. The pneumatic hammer of claim **1**, wherein the shaft is rotatably disposed through the first connecting hole, and the rigid rod is rotatable with the shaft.

5. The pneumatic hammer of claim **1**, wherein the working assembly further includes a nut, the driving member is detachably screwed to the rigid rod, and the nut is screwed to the rigid rod and abutted against the driving member.

6. The pneumatic hammer of claim **1**, wherein a plurality of ribs project on an inner surface of the main body, and the hammer body is slidably abutted against the plurality of ribs so that the hammer body and the inner surface of the main body are gapped.

7. The pneumatic hammer of claim **6**, wherein the main body is tubular, the main body includes a first section and a second section connected to each other, the first section and the second section define the interior space, the second section has an inner diametric dimension larger than an inner diametric dimension of the first section, and the plurality of ribs project on the second section.

8. The pneumatic hammer of claim **7**, wherein the second section has an outer diametric dimension larger than an outer diametric dimension of the first section, and the first section and the second section form a stepped structure.

9. The pneumatic hammer of claim **3**, wherein the shaft is rotatably disposed through the first connecting hole, and the rigid rod is rotatable with the shaft; the working assembly further includes a nut, the driving member is detachably screwed to the rigid rod, and the nut is screwed to the rigid rod and abutted against the driving member; a plurality of ribs project on an inner surface of the main body, and the hammer body is slidably abutted against the plurality of ribs so that the hammer body and the inner surface of the main body are gapped; the main body is tubular, two ends of the main body are in communication with outside, the main body includes a first section and a second section connected to each other, the first section and the second section define the interior space, the second section has an inner diametric dimension larger than an inner diametric dimension of the first section, and the plurality of ribs project on the second section; the second section has an outer diametric dimension larger than an outer diametric dimension of the first section, and the first section and the second section form a stepped structure; the handle assembly further includes a linkage and a lid, the linkage is laterally connected to and between the main body and the grip portion, the main body, the grip portion and the linkage are integrally formed of one piece, the lid covers an end of the main body remote from the working assembly, the lid includes a plurality of through holes, and the plurality of through holes are in communication with the interior space; the lid is made of rubber; the rigid rod has a length larger than a length of the main body; the rigid rod is a solid rod.

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